

Final Report

High-Performance Game AI for Nonaga: A Bitboard Engine with Minimax Search and Genetic Parameter Optimisation

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Summary

This project presents the design, implementation, and empirical evaluation of Nonaga-Online: a high-performance computational engine and adversarial AI agent for the abstract strategy board game Nonaga. Three core technical challenges are addressed. First, a custom hexagonal bitboard engine was implemented in C and exposed to Python via a Cython bridge, replacing an object-oriented pure-Python predecessor and achieving a $64\text{--}88\times$ improvement in nodes evaluated per second, sufficient to make real-time depth-3 adversarial search feasible for the first time. Second, an iterative-deepening minimax search with alpha-beta pruning and a Zobrist-hashed transposition table was developed, reducing the depth-3 node count from 82 million (brute force) to approximately 1.15 million, a combined $72\times$ reduction, and bringing decision latency from 49 seconds to under one second in steady-state play. Third, a Genetic Algorithm framework employing roulette wheel selection, arithmetic crossover, and a novel k -opponent circulant tournament fitness function was constructed to autonomously optimise the eight-parameter evaluation function. The resulting system supports real-time human-versus-AI gameplay through a PyGame interface and serves as a platform for continued heuristic experimentation. The project demonstrates that a custom bitboard architecture, even for a non-standard hexagonal geometry with no prior Nonaga-specific literature, can yield order-of-magnitude engine performance gains, and that principled adversarial search optimisation transforms an otherwise computationally intractable problem into one viable for interactive play.

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I would like to express my sincere gratitude to my supervisor, whose enthusiasm for this project was both consistent and genuine. Our conversations were substantive and wide-ranging, and his belief in the project's potential was a meaningful source of motivation throughout its development. I am also grateful to my assessor, whose critical observations during our meeting, in particular the advice to resist scope creep and commit to depth over breadth, materially shaped the direction of the final system.

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The theoretical foundations of this project, from algorithmic complexity to genetic computation, were built up over years of study, and this report would be entirely incomprehensible without that background. I am grateful for the opportunity to have applied that formation to a problem I found genuinely interesting.

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Chapter 1

Introduction and Background Research

1.1 Project Aim

The primary aim of this project is to architect and develop a high-performance computer program capable of playing the board game Nonaga at a highly competitive level. While playing a board game is intuitively simple for humans, teaching a machine to do so requires solving complex computational challenges.

To achieve this overarching aim, the project focuses on several core objectives, designed to tackle the problem from a high-level computational perspective:

- **Engine Optimization (The Board Representation):** Before an AI can play, it must understand the board efficiently. A major objective is to design a highly optimized internal representation of the game. This foundational layer must process rules and generate possible moves almost instantaneously, ensuring the AI can operate at high speeds without performance drag.
- **Adversarial Search (Looking Ahead):** The AI needs the ability to predict future outcomes. This objective involves implementing a strategic decision-making algorithm that allows the system to simulate thousands of potential future board states. By evaluating these outcomes, the system can quickly discard mathematically poor moves and select the most advantageous ones.
- **Evaluation Optimization (Developing Intuition):** The AI must be able to judge whether each board positions are advantageous. To implement this "intuition," the project implements an automated learning framework. By having the AI play thousands of simulated matches, the system systematically evolves and improves itself without relying on human intervention.
- **Interactive Client Development (User Interface):** To make the engine accessible and testable, the final objective is building a simple Graphical User Interface (GUI) that allows human users to play against the AI.

1.2 Introduction to Nonaga and the Computational Problem

1.2.1 What is Nonaga?

Nonaga is an abstract strategy board game created by Viktor Bautista i Roca that opposes two players playing as Red and Black [2]. The primary objective is straightforward: a player wins by manoeuvring their three individual circular pieces so that they connect and form a single continuous group as shown in Figure 1.1.

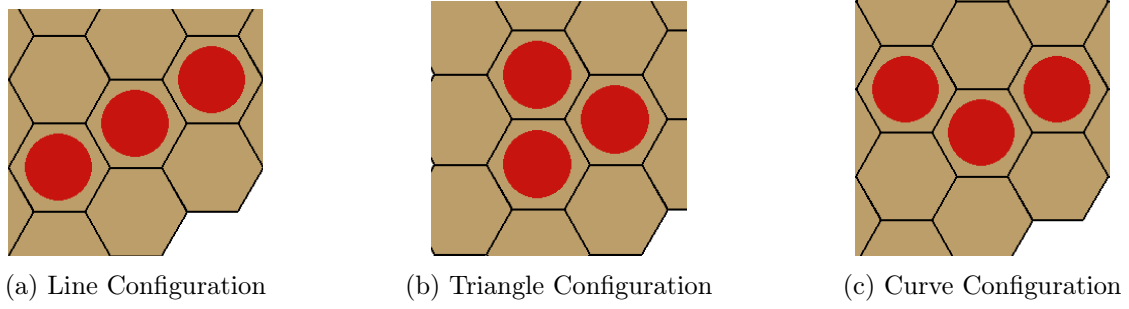


Figure 1.1: The three winning configuration shapes for Red.

The game is played on a dynamically shifting board composed of 19 independent hexagonal tiles. The structure of a turn consists of two distinct mandatory phases. First, a player must slide one of their pieces in a straight line; the piece cannot stop voluntarily and must continue moving until it collides with another piece or reaches the edge of the board (Figure 1.2a). Second, the player must alter the playing area itself by removing an unoccupied tile from the outer edge of the board and reattaching it to a different valid location on the perimeter (Figure 1.2b). An example game played between two human players can be found in [17]. Not all tiles can be moved in every position, the rules of tile movability are further explained in Appendix C.

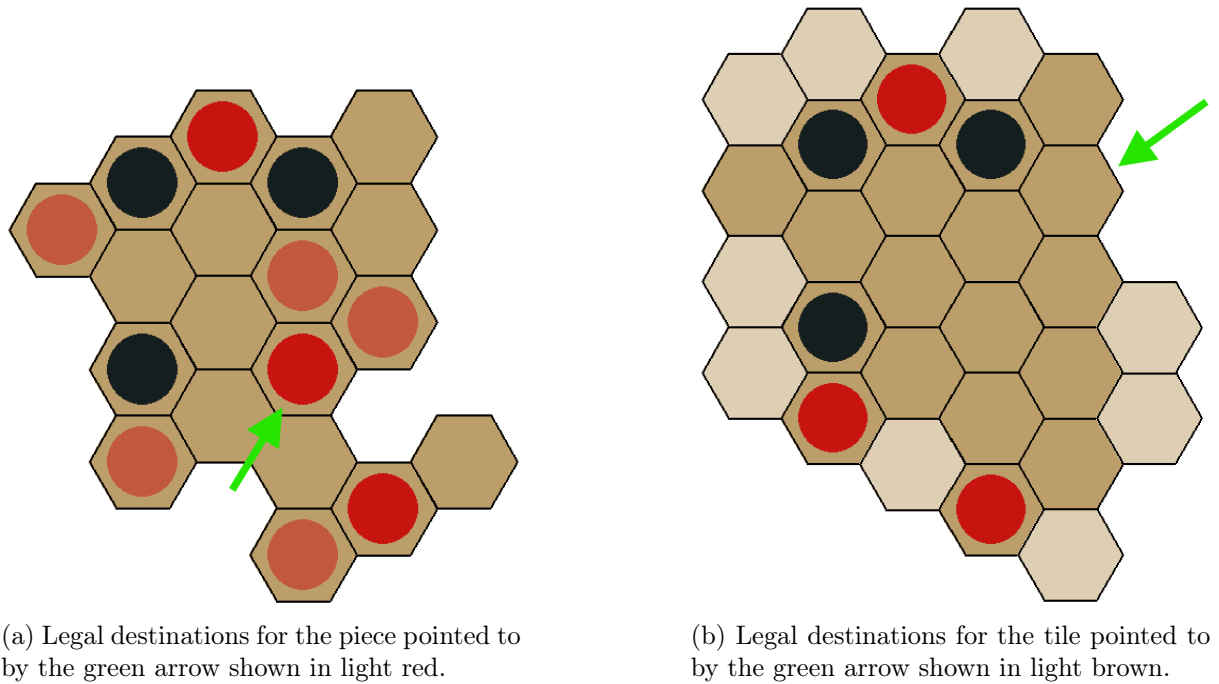


Figure 1.2: Piece and tile movement examples.

Although these mechanics are intuitive for a human player, this two-phase turn structure introduces severe computational complexity. Because a single turn requires both a piece movement and a tile movement which alters the structure of the board, the number of possible legal moves available at any given moment, is exceptionally high (estimated at 440 in Appendix D).

Moreover, the state of the board requires continuous recomputation; between each turn, the

legality of moving any specific tile must be dynamically evaluated based on the connectivity of its immediate neighbours.

Furthermore, to the best of our knowledge, there is currently no published academic literature addressing Nonaga.

1.3 Board State Representation

1.3.1 Hexagonal Coordinate Systems

Before establishing a board state representation, a fundamental design decision is the selection of a hexagonal coordinate system, as this choice directly governs the algebraic complexity of neighbour lookup, distance calculation, and direction traversal. Three primary systems are documented in the literature [25, 9]:

- Offset coordinates adapt the familiar square grid by shifting alternate rows or columns by half a cell width. While immediately intuitive, this system introduces an asymmetry: the direction vectors required to reach a neighbour differ depending on whether the current cell occupies an even or odd row, complicating the implementation of uniform geometric algorithms.
- Cube coordinates embed the hexagonal plane in three-dimensional space on the constraint surface $x + y + z = 0$, yielding uniformly simple distance and rotation formulae via unit-step axis permutations. The cost is a redundant third coordinate that must be carried and validated throughout all computations.
- Axial coordinates derive directly from cube coordinates by exploiting the linear constraint to eliminate the third axis, retaining only two values (q, r) . This preserves the uniform neighbour structure and constant direction vectors of the cube system while reducing the representational overhead, and produces a particularly clean mapping from hexagonal geometry onto linear memory structures [25].

1.3.2 Object-Oriented Representation

The most straightforward approach for board state representation is Object-Oriented Design (OOD), which instantiates discrete game entities as classes. While this method accelerates initial development and aligns with human conceptual models of a board game, OOD struggles to efficiently utilize computing resources in performance-critical applications [39]. Because objects are dynamically allocated, they become scattered across system memory. During the millions of rapid state evaluations required by an Artificial Intelligence (AI) search algorithm, this lack of spatial locality leads to severe CPU cache misses and memory latency, severely bottlenecking engine performance [1].

1.3.3 Bitboard Representation

On the opposite, modern high-performance engines rely on Data-Oriented Design (DOD), which prioritizes memory contiguity and hardware-level efficiency. By representing the game state as a series of 64-bit integers, known as "bitboards", the engine can map boolean states

(e.g., piece presence, color, or valid tiles) directly to individual bits. Studies have shown that contiguous, bit-level representations can achieve speedups of over 13x compared to equivalent OOP architectures by maximizing cache efficiency and allowing parallel evaluations via bitwise logical operations (AND, OR, XOR) [39].

While bitboards are the de facto standard in computer chess, they present unique implementation paradigms that must be carefully evaluated. Bitboard state transitions can theoretically operate 64 times faster than array encodings; however, poorly optimized bit-manipulation can introduce significant overhead, making the underlying mathematical model critical [32]. Furthermore, generalizing bitboards to non-standard geometries requires careful mathematical mapping [8].

Evaluating Bitboard Architectures for Sliding Pieces

A fundamental challenge in bitboard design is move generation, which is broadly categorized into "leaping" and "sliding" mechanics [8]. Leaping pieces (such as a Knight in chess) have static attack patterns relative to their coordinate, requiring a simple array lookup:

`Attacks[coordinate]`.

Conversely, sliding pieces, which move continuously along an axis until obstructed by the board edge or another piece, require dynamic calculation based on current board occupancy.

Historically, computing these sliding rays using step-by-step bitwise shifts was computationally expensive [15]. To mitigate this, two primary architectures used for chess have dominated engine design:

- **Rotated Bitboards:** An older, slower technique that maintains multiple redundant bitboards rotated along different axes (e.g., 90° , 45° , 315°). This transforms diagonal and vertical rays into contiguous horizontal memory segments, allowing fast array lookups at the cost of maintaining multiple synchronized board states [15].
- **Direct Lookup via Perfect Hashing (Magic Bitboards):** The modern, memory-intensive, state-of-the-art methodology [37]. This technique isolates the pieces along a slider piece's ray (blocking their path) and multiplies this occupancy bitboard by a pre-calculated 64-bit integer (the "Magic Number"). Following a bit-shift, this operation yields a dense, perfect hash index used to fetch the bitboard corresponding to the coordinates visible to the sliding piece. This reduces complex ray-casting and collision detection to a handful of CPU cycles: `Mask` $\rightarrow$ `Multiply` $\rightarrow$ `Shift` $\rightarrow$ `Array Lookup`.

1.4 Adversarial Search

Nonaga is formalized as a two-player, deterministic, zero-sum game of perfect information. Identifying optimal strategies within such environments traditionally necessitates traversing the state-space via adversarial search algorithms. Currently, the two pre-eminent paradigms for state-space evaluation in game artificial intelligence (AI) are Monte Carlo Tree Search (MCTS) and the Minimax algorithm [10].

1.4.1 Monte Carlo Tree Search

MCTS has demonstrated profound empirical success in modern game AI, particularly within domains characterized by massive branching factors, such as Go [10]. The algorithm evaluates game states by incrementally constructing an asymmetric search tree and simulating thousands of randomized playouts (rollouts) to terminal states. A distinct advantage of MCTS is its ability to estimate state values without relying on complex, domain-specific heuristic evaluation functions. Conversely, MCTS imposes substantial memory overhead to maintain the search tree, and the stochastic nature of its rollouts necessitates considerable computational time to converge on statistically robust estimations [27].

1.4.2 Minimax

Minimax is a foundational game-theoretic algorithm that performs a recursive, depth-first exploration of the game tree. In its theoretical form, searching exhaustively to terminal nodes, it guarantees the computation of a deterministic Nash equilibrium [27]. However, naive Minimax rapidly becomes computationally intractable for games with substantial branching factors (b) and depths (d), exhibiting an exponential time complexity of $O(b^d)$ [29]. Consequently, practical implementations are restricted to a constrained depth limit, substituting terminal game states with a static, expert-crafted heuristic evaluation function [33]. It is critical to note that under these constraints, the algorithm no longer guarantees a true Nash equilibrium; rather, it computes an approximated optimal strategy. The reliability of this approximation is strictly bounded by the fidelity of the heuristic function, making the engine susceptible to the horizon effect if the heuristic incorrectly evaluates the true strength of a leaf node [29]. To mitigate the exponential explosion of the search tree and allow for deeper, more accurate heuristic evaluations, Minimax is typically augmented with algorithmic enhancements.

Alpha-Beta Pruning

Alpha-Beta pruning is an optimization technique that mathematically guarantees the return of the identical optimal decision as standard Minimax, but dramatically reduces the requisite number of node evaluations [20]. It achieves this by maintaining upper and lower bounds (alpha and beta) and discarding branches mathematically proven to yield worse outcomes than previously evaluated paths. In the best case, Alpha-Beta pruning effectively halves the exponent of the algorithmic complexity from $O(b^d)$ to $O(b^{d/2})$, yielding an effective branching factor of $\sqrt{b}$ [29]. In the worst case, it evaluates as many nodes as standard Minimax [20]. In highly optimized implementations, advanced engines leverage this method to reduce their effective branching factor to nearly 1.5, permitting significantly deeper searches within identical time constraints [10].

Transposition Table

A Transposition Table (TT) is a dynamic hash table utilized to cache the heuristic evaluations and search depths of previously visited game states [23]. Due to the overlapping nature of game trees, identical states can often be reached via different permutations of moves (transpositions). The TT allows the algorithm to bypass redundant subtree recalculations, fundamentally

enabling efficient Iterative Deepening [36]. To minimize the computational cost of state serialization and mitigate hash collisions, the TT typically employs Zobrist Hashing, generating a near-unique integer for a given board configuration via XOR operations [40]. However, the overhead associated with TT maintenance, specifically state hashing, memory access latency, and collision resolution, must be critically balanced [7]. Poorly optimized TT maintenance overhead can exhibit performance degradation that eclipses the computational benefits of pruning depending on the nature of the search space and specific implementation.

Move Ordering and Iterative Deepening

Move Ordering is a heuristic technique that dictates the sequence in which child nodes are generated and evaluated by the Alpha-Beta algorithm [29, 5]. The primary objective is to evaluate the strongest candidate moves first, establishing rigid alpha and beta bounds early in the search, thereby maximizing the frequency of subsequent node cutoffs [5]. This enhancement is highly synergistic with Iterative Deepening, which conducts sequential depth-limited searches. By caching the principal variation (the sequence of best moves) discovered at depth $i - 1$, the algorithm can prioritize exploring these nodes first at depth i [31, 6]. Additionally, the ordering of sibling nodes can be achieved through either static domain-dependent evaluations or dynamic search-based heuristics. While computationally inexpensive static evaluations can be applied to rank nodes based on game-specific features (such as capture priority), they introduce algorithmic overhead. To bypass this computational bottleneck, advanced engines heavily rely on dynamic, domain-independent techniques such as the Killer Heuristic and the History Heuristic [29]. These methods prioritize moves that have successfully provoked beta-cutoffs in parallel branches of the search tree, exploiting the localized relevance of tactical threats. The efficacy of Move Ordering is intrinsically linked to the accuracy of these combined techniques; if the prescribed order fails to reflect the true optimal sequence, the engine incurs sorting and evaluation overhead without the corresponding pruning benefits. Consequently, the computational cost of the ordering mechanism must remain strictly minimal to ensure a net-positive performance yield [5, 6].

1.5 Heuristic Evaluation and Parameter Optimization

Because exhaustive search to terminal nodes is computationally intractable for games with substantial branching factors, the Minimax algorithm must terminate at a constrained depth limit. At these leaf nodes, a static heuristic evaluation function is deployed to approximate the true game-theoretic utility of the state.

Linear Evaluation Functions

The foundational approach to static evaluation is the linear evaluation function [33]. This model quantifies a game state by extracting a vector of discrete, domain-specific features (e.g., piece mobility, structural connectivity, or center control) and calculating their weighted sum. Mathematically, it is expressed as $E(s) = \sum_{i=1}^n w_i f_i(s)$, where f_i represents the extracted feature and w_i is its corresponding static weight. The primary architectural advantage of a linear function is its extreme computational efficiency. Because it relies exclusively on scalar

multiplication and addition, it introduces minimal overhead to the engine, maximizing the nodes-per-second (NPS) throughput. This high-speed evaluation is highly synergistic with Alpha-Beta pruning, allowing the search algorithm to reach significantly deeper horizons within identical time constraints [31, 5]. However, its fundamental limitation is its inability to natively model non-linear interactions; if the strategic value of one feature is strictly dependent on the presence of another, a purely linear combination cannot inherently capture this dependency [35, 22].

Neural Networks and Non-Linear Models

To address the limitations of linear models, modern game-playing architectures leverage Deep Neural Networks (DNNs) to evaluate board states. As demonstrated by deep reinforcement learning systems like AlphaZero [35], DNNs can successfully approximate highly complex, non-linear relationships between board features [14], yielding superhuman evaluative accuracy. However, this accuracy incurs a massive computational penalty. The requisite tensor operations and matrix multiplications are exceptionally slow compared to basic arithmetic, dramatically reducing the maximum depth a purely CPU-bound Minimax engine can reach [35]. To bridge this computational divide, state-of-the-art engines frequently employ Efficiently Updatable Neural Networks (NNUE) [22]. NNUEs utilize shallow network architectures where only the specific neurons affected by a move are recalculated [22], offering a compromise between the rich, non-linear accuracy of a neural network and the necessary execution speed of a traditional linear heuristic.

1.5.1 Automated Parameter Tuning

While linear evaluation functions offer optimal search speeds, their efficacy is entirely predicated on the configuration of their weight vector $(w_1 \dots w_n)$ [33]. Because manual tuning relies on inherently subjective domain expertise and becomes impractical in multi-dimensional spaces, discovering the optimal weights is formalized as a continuous or discrete optimization problem. Various automated paradigms exist to navigate this search space.

Hill Climbing

Local search metaheuristics, such as Hill Climbing, iteratively perturb individual weights and adopt them if benchmarks yield higher win rates. While computationally lightweight, this method is notoriously susceptible to premature convergence, frequently trapping the optimizer in local optima within the complex fitness landscapes of game-tree heuristics [29, 5].

Simulated Annealing

Simulated Annealing (SA) [19] extends Hill Climbing by incorporating a probabilistic acceptance mechanism inspired by the physical annealing process in metallurgy. Rather than exclusively adopting improvements, SA accepts worse-performing weight configurations with a probability governed by a temperature parameter, which decreases monotonically according to a cooling schedule. This enables the optimizer to escape local optima that trap purely greedy methods, offering a more thorough exploration of the fitness landscape [29]. However, the

quality of convergence is critically sensitive to the design of the cooling schedule; an excessively rapid cooling rate risks premature convergence analogous to Hill Climbing, while an overly gradual schedule renders the process computationally prohibitive [19].

Temporal Difference

Alternatively, Temporal Difference (TD) learning updates weights dynamically based on the temporal sequence of game states, using gradient descent to minimize the error between consecutive evaluations. This method, famously utilized in Arthur Samuel’s seminal checkers program [30], allows the engine to learn incrementally from continuous self-play without requiring explicit datasets. However, TD learning requires mathematically rigid formulations of the reward signal and learning rate; which if misconfigured can become highly unstable [35].

Standard Evolutionary Algorithms

Genetic Algorithms (GAs) [13] provide a robust, population-based alternative for global optimization. Inspired by biological evolution, a standard GA maintains a diverse population of candidate weight vectors (chromosomes). Through successive generations, these vectors are evaluated via fitness functions, typically structured as round-robin or Swiss-system tournaments, where the highest-performing configurations are selected for reproduction [34]. By applying stochastic operators such as crossover (recombining weights from two parents) and mutation (randomly altering specific weights), GAs excel at broad exploration of the search space, mitigating the local optima traps that plague local search methods [18]. Despite their exploratory strengths, standard GAs exhibit structural limitations when optimizing highly correlated parameters. In many board games, heuristic features are interdependent; for example, the optimal weight for piece mobility may strictly depend on the weight assigned to board centrality. This phenomenon, known as epistasis [13], can cause standard, position-independent crossover operators to continuously destruct highly effective combinations of weights (building blocks) during reproduction, severely bottlenecking the algorithm’s optimization trajectory.

Linkage Learning in Genetic Algorithms

To address the destructive nature of standard crossover operations in epistatic environments [13], modern evolutionary computation introduces Linkage Learning [24]. This advanced paradigm augments the traditional GA by actively identifying dependencies and correlations between specific genes within the chromosome. Instead of relying on blind, uniform crossover, Linkage Learning algorithms often employ underlying machine learning models or statistical dependency matrices [26] to map the structure of high-performing genetic building blocks. During reproduction, these linked clusters of weights are preserved and transferred as a single unit, ensuring that synergistic parameter interactions are maintained across generations. While Linkage Learning significantly improves the quality and robustness of the generated offspring in complex search spaces, it introduces substantial algorithmic complexity and computational overhead to the generational update phase, requiring rigorous mathematical profiling to ensure viability [38].

Chapter 2

Methodology

2.1 Core Engine Design: Representation and Computation

Given Nonaga’s branching factor of 440 (established in Appendix D), reaching a search depth of 3–4 within a human-playable time budget of a few seconds per move demands a game engine capable of evaluating tens of thousands of nodes per second. Every architectural decision in this chapter is evaluated against this core computational constraint.

2.1.1 Board State Representation

A fundamental design choice in the development of the game engine was the internal representation of the Nonaga board. Traditional implementations of hexagonal board games often rely on 2D arrays or object-oriented graph structures. However, these approaches incur significant overhead and CPU cache-miss penalties during the millions of node evaluations required by deep adversarial search.

To achieve maximum computational throughput, a Bitboard architecture was implemented. By mapping the 19-cell hexagonal grid of Nonaga to an array of 64-bit integers, the entire game state is compressed. Consequently, complex operations such as move generation, tile movability status, and win-condition validation are reduced to highly efficient, CPU-native bitwise operations (AND, OR, XOR) that evaluate the board state in parallel against precomputed bitmasks. While the application of bitboards is well-documented for standard board geometries [37] and has been explored for non-standard ones [8], adapting this architecture to Nonaga’s unique, dynamic topology represented a calculated engineering risk, as no prior work addressed a hexagonally-restructuring game board specifically. The primary risk was an increase in development time and architectural complexity, with the contingency that the geometric regularity of the hexagonal grid would permit a systematic derivation of the required bitmasks.

2.2 Artificial Intelligence and Search Strategy

2.2.1 Adversarial Search Algorithm

While Monte Carlo Tree Search (MCTS) is a popular choice for games with high branching factors, an enhanced Minimax algorithm was selected for this project. Because Nonaga is heavily geometric, its strategic value is largely expressible through spatial features such as piece proximity and axial alignment, which are directly computable from the bitboard state, allowing a highly accurate heuristic evaluation function to be reliably constructed. Leveraging this domain knowledge via Minimax was estimated to provide a more deterministic and robust early-game performance than the randomized rollouts of standard MCTS, a conclusion supported by comparative studies in domains where reliable heuristics can be constructed [10, 27].

The primary decision-making agent utilizes this Minimax algorithm, significantly optimized via Alpha-Beta Pruning, Iterative Deepening, and a Transposition Table (TT) backed by Zobrist Hashing. Furthermore, the TT does more than just eliminate the redundant computation of identical board states; it serves as a highly efficient First-Move Ordering mechanism maximizing the efficiency of the Alpha-Beta pruning process. Instead of incurring the heavy computational overhead of dynamically sorting all possible legal moves at every node, the engine utilizes a Hash Move heuristic. By saving the hash of the best move encountered during the previous Iterative Deepening depth and retrieving it via the TT to evaluate first, the engine captures the vast majority of pruning benefits with a fraction of the algorithmic complexity.

2.2.2 Evaluation Function

Because terminal win-states cannot be reached at shallow depths, the Minimax algorithm relies heavily on a parameterized heuristic evaluation function. This naturally integrates with Nonaga’s highly geometric nature. Each parameter can be multiplied to a geometric feature of the board that can be efficiently derived from the bitboard. A linear evaluation function [33] is particularly well-suited to this context, as it is computationally inexpensive to evaluate at every node of a deep search tree, whereas non-linear methods such as neural networks would impose a per-node latency incompatible with the depth required for competent play. Given Nonaga’s significant branching factor, development resources were furthermore prioritised towards maximising search depth rather than evaluation complexity.

2.3 Evaluation Function Optimization

2.3.1 GA Framework Design and Operator Selection

To systematically optimize the weights of these parameters, a Genetic Algorithm (GA) framework was selected over traditional hill-climbing methodologies. The weights of the evaluation function are mutually dependent: the optimal value of one parameter is contingent on the values of the others. This interdependency creates a fitness landscape susceptible to premature convergence in gradient-following methods, making a population-based approach a more robust alternative [13].

A Random K-Opponent Tournament Scheduler was engineered, forcing genomes to compete against each other in self-play to determine their fitness. A unique challenge in optimizing heuristics for adversarial games is that fitness is entirely relative; a gene’s performance is strictly dependent on the quality of its opponents. To mitigate this noise, the K parameter (number of opponents) must be set as high as computationally acceptable to ensure a statistically significant fitness score. In practice, K was determined empirically by incrementally increasing the number of opponents until the variance in fitness rankings across repeated runs stabilised. Roulette wheel selection and arithmetic crossover were selected as the GA operators, as these are well-established for continuous-valued parameter optimisation [18]; their concrete implementation is detailed in Section 3.3.

It was recommended to benchmark the GA’s convergence against an absolute "apex gene" found via brute-force search over a restricted domain (1–6 per parameter) [16]. However,

calculations revealed this to be unfeasible due to the estimated execution time and heavily neighbourhood-dependent nature of the fitness landscape (further detailed in Appendix E). While advanced evolutionary strategies, specifically Linkage Learning, were proposed as a potential solution to this landscape dependency [16], their implementation was ultimately deemed out of scope. The core objective of the GA was to tune the evaluation function parameters to a highly performant standard, which the implemented baseline achieved successfully (as discussed in Section 4.3). For that reason, development resources were strategically reallocated to engine polishing and system optimization rather than over-engineering the evolutionary framework.

2.4 System Architecture and Technology Stack

2.4.1 Language Selection and Hybrid Integration

The overarching architectural goal was to deliver an interactive desktop client without compromising the extreme execution speeds required by the AI engine. Pure Python, while excellent for rapid UI prototyping via the PyGame library, was deemed fundamentally unsuitable for the core algorithmic engine. The Python Global Interpreter Lock (GIL) and the inherent overhead of dynamic object handling introduce severe bottlenecks when executing computationally dense bitwise operations and recursive tree searches [3].

To resolve this, a hybrid technology stack was engineered. The low-level Bitboard logic and Minimax core were strictly implemented in standard C, granting precise control over memory allocation and maximum instruction optimization. Python was retained exclusively for high-level orchestration, user interface rendering, and the GA tournament scheduling. Cython was utilized as the critical bridging layer, allowing the C-compiled backend to be exposed as native Python modules [4]. This multi-language architecture cleanly separates the presentation layer from the algorithmic core, guaranteeing that both the interactive gameplay and the offline GA training pipelines execute against the exact same high-performance ruleset without rule drift or latency.

C was preferred over managed-runtime alternatives such as Java for the performance-critical kernel, as it avoids garbage collection pauses and provides the predictable, cache-friendly memory access critical for tight search loops [39, 1]. C++ was considered but rejected, as its additional abstraction overhead offered no measurable advantage over C for this narrowly scoped kernel. Existing proficiency with both Python and C further reduced development overhead and integration risk. The primary risk of this approach was the increase in development time and debugging complexity inherent to C, which lacks the high-level abstractions available in Python; this was mitigated by strictly limiting C code to the well-defined bitboard and search kernel, with Python retained for all higher-level logic. The resulting component structure of the main game client, illustrating which classes invoke one another and the rationale for those dependencies, is shown in Figure 2.1.

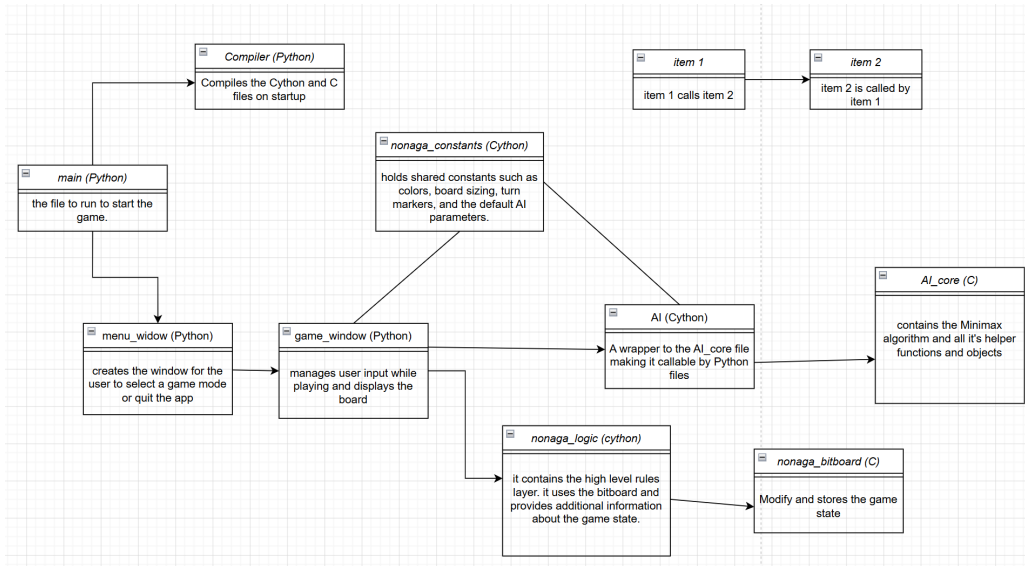


Figure 2.1: Class diagram of the main game client, illustrating the call dependencies between core software components.

2.5 Project Management

Throughout the project lifecycle, `git` was employed as the primary version control system to manage the evolving codebase, track granular changes, and maintain a secure repository.

Development was structured around a feature-branch workflow, with each phase's deliverable developed and stabilised on a dedicated branch before integration into the main trunk. This ensured that regressions introduced during low-level optimisation (Phase 3) could be isolated without destabilising the working implementation from earlier phases.

Initially, a rigid, linear project plan was formulated in the Project Outline and Specification; however, early implementation phases revealed that this structure was overly abstracted and practically restrictive. The original plan operated on the flawed assumption of strict modular independence. In reality, the high degree of uncertainty in the exact implementation methods rendered the strict Waterfall approach implied in the plan non-viable.

Consequently, the project management strategy shifted to an adaptive, feature-driven Agile methodology. Development followed a verification-first discipline: each feature was considered complete only once it had been validated against a defined set of functional criteria, with custom diagnostic tooling developed specifically to support this process (detailed in Section 3.4.1). The workflow followed a strict iterative loop: a target feature was engineered, rigorously verified against its functional requirements, and fully stabilized before continuous research dictated the selection and prioritization of the subsequent development target.

2.5.1 Iterative Development Phases

Because the project required continuous research and adaptation, the development lifecycle was structured around flexible objectives rather than strict time-boxed intervals. Each phase focused on delivering a functional, verifiable component of the overarching system. These phases map directly to the four project objectives outlined in Section 1.1: Phase 1 delivers the interactive GUI, Phases 2 and 3 deliver the high-performance engine and adversarial search,

Phase 4 delivers the validation infrastructure, and Phase 5 delivers the GA optimisation pipeline.

- **Phase 1: Core Mechanics and Visualization:** The initial focus was establishing a ground-truth representation of the game. The core rules of Nonaga were implemented in Python, paired with an interactive PyGame graphical user interface. By the end of this phase, human players could execute a valid game from start to finish, establishing a reliable baseline for future AI integration.
- **Phase 2: Adversarial Search Integration:** With a valid state representation in place, the focus shifted to algorithmic decision-making. By the end of this phase, the Minimax algorithm, enhanced with alpha-beta pruning, was successfully implemented to evaluate game trees.
- **Phase 3: Low-Level Optimization:** The objective of this phase was to minimize execution latency, thereby incrementally increasing the permissible search depth of the AI. By the end of this phase, the AI achieved a stable search depth of 3 within a pragmatic execution timeframe (a few seconds per move). This phase encompassed the language migration of performance-critical loops from pure Python to Cython/C, alongside the integration of bitboards, Transposition Tables, and move-ordering heuristics.
- **Phase 4: Diagnostic Tooling and Verification:** Custom diagnostic UI tools (detailed in Section 3.4.1) were developed to manually verify the functional validity of the underlying engine modifications. This phase was introduced to establish a robust debugging infrastructure, facilitating the immediate isolation of logical regressions and minimizing subsequent debugging overhead as the architecture grew in complexity.
- **Phase 5: Evolutionary Tuning and Profiling:** The final phase focused on maximizing the AI's heuristic accuracy and computational efficiency. By the end of this phase, a modular Genetic Algorithm framework was constructed to automate the tuning of evaluation weights, resulting in a highly proficient artificial opponent.

This phase-driven approach provided the necessary flexibility to adapt to emergent technical challenges. Rather than adhering strictly to predetermined milestones, development iteratively cycled back to previous phases to address critical bottlenecks such as returning to core optimization (Phase 3) when subsequent heuristic tuning (Phase 5) revealed underlying performance constraints. Ultimately, the priority was to engineer a robust, high-performing AI, justifying deviations from the initial, rigid project specification.

2.5.2 Scope Management

As advised during the interim progress review, a pragmatic approach to scope management was adopted [16]. Rather than expanding the project horizontally to encompass distributed cloud computing or widespread multi-platform deployment, development deliberately prioritized the deep optimization and polishing of the existing core architecture. This targeted focus ensured that the underlying AI engine and the local tuning frameworks achieved a rigorous standard of performance and stability without stretching the project's resources too thin.

Chapter 3

Implementation and Validation

3.1 The High-Performance Game Engine (C/Cython)

3.1.1 Coordinate System

As outlined in Section 1.3.1, three hexagonal coordinate systems are available, each with distinct trade-offs. The engine adopts the axial coordinate system (q, r) as its internal spatial representation. Axial coordinates provide uniform direction vectors and branch-free neighbour arithmetic, both of which are essential properties for the high-frequency geometric queries demanded by move generation and tile legality checking. Crucially, the two-axis representation admits a direct, linear mapping from any (q, r) pair to a one-dimensional bit index, avoiding the conditional row-parity corrections that an offset representation would impose at every neighbour lookup. The precise derivation of this mapping is detailed in the following subsection.

3.1.2 Bitboard Representation: Memory and Architecture

Unlike traditional board game implementations that use Object-Oriented coordinate structures, this engine relies on a "bitboard" architecture. In this paradigm, the board is not stored as a collection of objects; instead, the hexagonal play space is flattened, and every single hexagonal cell is mapped to a single bit (0 or 1) inside a standard 64-bit integer as seen in Figure 3.1a. In the integer used to store the board, each bit represents a certain coordinate. In a game like chess, where the board is static, this integer contains one bit for each square on the board. Doing the same for Nonaga, where the storing integer contains as many bits as there are tiles that form the board, would fail given the dynamic nature of the Nonaga board. To fix this issue, the integer is set to have an arbitrary number of bits such that it can express a large array of positions in which the whole Nonaga game will be played which will be referred to as the play-field. The play-field was chosen to be a hexagon of radius 10, containing 441 cells, so that it's size doesn't limit game-play (Illustrated in Figure 3.1b). Conveniently, the entire board state can be encoded using an array of seven 64-bit integers ($7 \times 64 = 448$). Since the play-field is finite it theoretically bounds the number of possible moves, in the event that the board drifts to one of its edges. In practice, this situation is very unlikely to happen through intelligent or random play and was never encountered through all the games simulated during the development of this project.

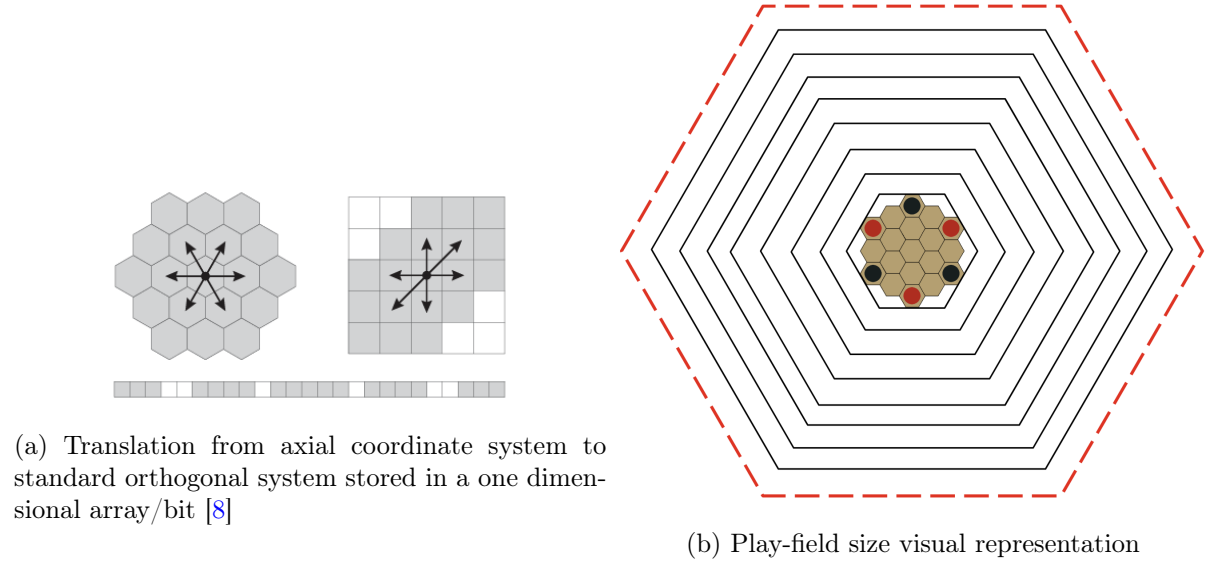


Figure 3.1

To map the hexagonal board geometry into a linear bit sequence, the engine employs a coordinate transformation from an axial hex-coordinate system (q, r) to a flattened one-dimensional index [8]. Because axial coordinates can be negative, a constant spatial offset (`BOARD_OFFSET = 10`) is applied to shift the origin, ensuring all values fall within a non-negative range. The hexagonal space is subsequently embedded into a 2D square grid defined by `BOARD_WIDTH = 10 + 1 + 10 = 21`. The linear index is then computed using row-major flattening:

$$\text{index} = (q + 10) + (r + 10) \times 21$$

This index directly targets a specific bit within the 448-bit span. To locate this bit in the array of seven 64-bit integers, the engine utilizes rapid bitwise operations: $\text{index} \gg 6$ (right shift by 6 to divide by $64 = 2^6$) yields the 64-bit word index, while $\text{index} \& 63$ (bitwise AND with 63 equivalent to modulo 64) isolates the exact bit position within that word.

To illustrate this translation, consider a tile at the axial coordinate $q = -2, r = 0$. Applying the offset yields the non-negative coordinates $q_{\text{shifted}} = 8$ and $r_{\text{shifted}} = 10$. The row-major flattening then calculates a linear index of $8 + (10 \times 21) = 218$. Utilizing the bitwise operations, the engine determines the word index ($218 \gg 6 = 3$) and the bit position ($218 \& 63 = 26$). Thus, the cell's state is stored precisely at the 26th bit of the 4th integer (index 3) in the bitboard array.

Conversely, to retrieve spatial geometry from a set bit, an inverse mapping is applied. The index is derived from the word and bit position:

$$\text{index} = (\text{word_index} \times 64) + \text{bit_position}$$

The (q, r) axial coordinates are extracted via modulo operations ($\text{index} \bmod 21$) and integer division ($\lfloor \text{index} / 21 \rfloor$) respectively. This seamless bidirectional mapping strictly relies on primitive integer arithmetic, avoiding the overhead of complex data structures while accurately preserving hexagonal topology as illustrated in Figure 3.1a.

Because a single bit can only represent a binary state (e.g., "occupied" or "empty"), the

complete game environment cannot be modelled within a single flat array. Instead, the engine maintains distinct bitboards representing different logical layers of the game: red pieces, black pieces, all tiles, movable tiles, unmovable tiles, and a cache of valid tile destinations.

- **red pieces:** an array of 7×64 bits in which a 1 corresponds to a red piece being present at the corresponding coordinate.
- **black pieces:** the same as red pieces but for black pieces.
- **all tiles:** an array of 7×64 where each 1 corresponds to a tile being present at the corresponding coordinate.
- **movable tiles:** a state representation where a bit value of 1 identifies a tile structurally free to be translated without violating Nonaga's rules.
- **unmovable tiles:** the disjoint complement to movable tiles under all tiles, tracking pieces that are currently pinned by surrounding geometry and blocked from movement.
- **valid tile destinations:** a continuously updated cache where set bits map to all legally permissible landing coordinates across the current board topology.

The primary advantage of this architecture is that game logic is no longer processed through costly memory allocations or nested loops over data structures. Instead, state evaluation is executed using hardware-sympathetic operations: flipping, testing, shifting, and scanning bits directly at the CPU level.

3.1.3 Optimized Move Generation Algorithms

To support the heavy computational demands of deep game tree searches (such as Alpha-Beta pruning), the engine's move generation must minimize CPU cycles. The engine achieves this through optimization layers, a few of which will be explained here.

Bare Metal Legality Check

Validating whether a single cell is a legal landing destination (`candidate_is_legal_for_tiles`) is the most frequently executed function in the engine. To optimize this, the engine avoids traditional spatial pathfinding entirely. Legality of a tile move in Nonaga relies solely on the immediate topology of a cell (Appendix C). Therefore, the engine only evaluates the six immediate neighbours of a candidate cell. The engine performs a mask to ask a binary question: "Which neighbours are occupied?" The answers for these six neighbours are packed into a single 6-bit integer (representing a value from 0 to 63). This integer is then used as a direct index into a precomputed 64-entry lookup table containing all legal local shapes. Because the complex geometric rules of the game are written into this lookup table at compilation, validating any candidate cell resolves in a few CPU instructions.

Sparse Traversal and Global Caching

While movable tiles possess distinct valid destinations (Figures 3.2 and 3.3), there is typically a substantial intersection between their respective move sets.

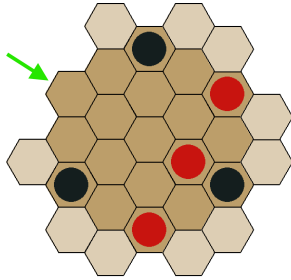


Figure 3.2: The possible move set for the selected tile.

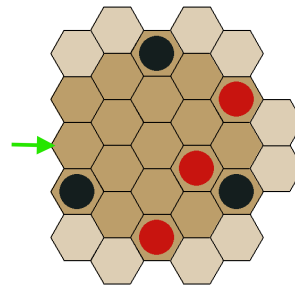


Figure 3.3: The possible move set for an alternative selected tile.

To exploit this, the engine constructs a global cache of all theoretically legal destinations (`rebuild_tile_destination_cache`). Rather than iterating through all 448 cells one-by-one, where the vast majority are empty space, the engine uses sparse bitboard traversal. By leveraging "find next set bit" hardware operations, the CPU jumps directly from one occupied tile to the next.

For each occupied tile found, its adjacent empty cells are recorded in a temporary bitset. Because multiple tiles might point to the same empty neighbour, writing this to a bitset inherently deduplicates overlapping cells. Only these specific, pre-filtered candidates are then passed through the bare metal legality check.

Differential Move Generation

When the engine needs to query the legal moves for one specific tile (`bitboard_get_valid_tile_positions_for_tile`), it employs a highly efficient differential update strategy.

Removing a source tile from the board logically alters the overall geometry. However, because legality is strictly dependent on immediate neighbours, removing a tile only affects the legality of its six immediate neighbours. Any cell farther away still possesses the exact same neighbour pattern as before.

The engine capitalizes on this mathematical guarantee by combining two methods. For any cell outside this 6-cell affected area, the engine completely trusts the global destination cache, instantly retrieving valid moves. The engine then distrusts the cache and re-runs the bare metal legality check only for the cells within the 6-cell subset. By utilizing the global cache for the majority of the board and only paying the computational cost for local rechecks near the moved tile, the engine achieves an exceptionally high work-per-instruction ratio.

3.2 AI and Search Implementation

3.2.1 Zobrist Hashing for Transposition Table

To optimize the Minimax search and handle the convergence of different move sequences into identical board states, a Transposition Table (TT) was implemented (Section 1.4.2). Because the Nonaga game state is compactly represented using a bitboard architecture (Section 3.1), mapping it into a unique 64-bit "fingerprint" using Zobrist hashing is highly efficient. The process begins by initializing a table of pseudo-random 64-bit integers generated from a deterministic seed. Each potential feature of the game state (e.g., a red piece, black piece, or tile occupying a specific cell index) is assigned a unique random key. The global hash for a given position is then computed by applying the bitwise XOR operator to the keys of all present features. This approach is computationally efficient due to the order-independent property of XOR; the engine iterates through the bitboard's seven 64-bit words, identifies set bits, and accumulates the corresponding keys without requiring expensive re-calculations.

3.2.2 Transposition Table Integration and Best Move Ordering

Retrieval

During evaluation, the recursive `ai_minimax_piece` and `ai_minimax_tile` routines probe the TT before enumerating and expanding candidate nodes. Upon a hit, the engine extracts the cached depth, evaluation score, bounding flags (`EXACT`, `LOWER`, `UPPER`), and historical best moves, rigorously validating them against the current board to prevent hash collision errors. If a valid entry meets the required search depth, it triggers an immediate cutoff: either returning exact evaluations directly, or tightening the search window ($\alpha \geq \beta$) using bound resolutions.

Writing

Upon completing a node, the engine writes its computed Zobrist hash, depth, heuristic value, bound flag, and best moves back into the indexed slot.

Best Move Ordering

This mechanism acts as the backbone for the AI's classical Iterative Deepening framework. Implemented as a fixed loop from depth 1 up to `max_depth` (rather than a time-budgeted variant), each iteration resets the root state, runs a full minimax search, and overwrites the overall best move. This takes advantage of the TT and allows Best Move Ordering.

3.2.3 Parameterized Evaluation Function

The evaluation function utilizes a total of eight features and tuned weights to evaluate a position. Four weights are applied to features extracted from the current player's piece configuration, while the remaining four are applied to the opponent's configuration. The total heuristic value of a state S is the difference between the player's features and the opponent's, formalized below.

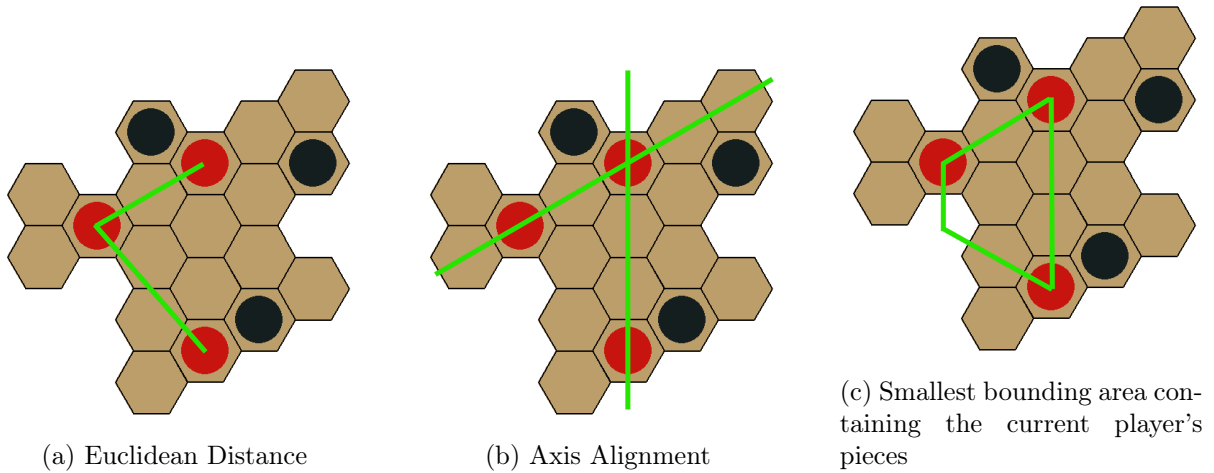


Figure 3.4: Visual support to the four heuristic features evaluated by the evaluation function.

1. **Euclidean Distance:** As illustrated in Figure 3.4a, this feature quantifies the proximity of a player's pieces. Rather than calculating the total distance between all piece pairs, the heuristic sums only the two smallest distances. This approach accurately reflects the game's win condition: pieces only require a single adjacent neighbour to form a continuous chain. Minimizing these two specific distances directly facilitates the formation of a win-state without requiring redundant proximity between all three pieces.
2. **Axis Alignment:** This feature, visualized in Figure 3.4b, measures the degree to which a player's pieces share a common hexagonal axis. In the game's mechanics, pieces move in straight lines until an obstacle is reached; therefore, aligning pieces on the same axis is a critical strategic precursor to forming a connection. The heuristic assigns a value (0, 1, or 2) based on the number of collinear pairs. Notably, even if all three pieces are mutually aligned on a single axis, the maximum value remains 2, as this represents the minimum number of connections required to link three discrete points into a single continuous chain.
3. **Missing Tiles:** This feature quantifies the "missing" space within the smallest bounding area containing a player's pieces (The area can be seen Figure 3.4c). In a dynamic hexagonal grid, empty spaces function as impassable voids that disrupt movement connection paths. A higher count of missing tiles increases the "topological distance" between pieces, forcing the agent to execute more complex, multi-turn manoeuvres to bridge the gaps. This forces the evaluation function to prioritize board configurations that maintain a solid, manoeuvrable surface.
4. **Opponent Disruption:** Similarly to Feature 3, this feature counts the number of opponent pieces in the smallest bounding area containing a player's pieces (The area can be seen Figure 3.4c). A higher number of opponent pieces between the current player's pieces makes it easier for them to impede the current player's plans.

3.3 The Genetic Algorithm Framework

3.3.1 GA Architecture and Encoding

The encoding strategy maps directly to the linear evaluation function, which requires eight distinct parameters. Consequently, a chromosome is represented as a continuous-like integer array of size eight, bounded within the interval $[0, 50]$. Each gene corresponds to a specific weight within the evaluation function.

3.3.2 Fitness Evaluation via Tournaments

To evaluate the fitness of each chromosome, a tournament-style selection process is employed. An AI agent equipped with a specific chromosome's weights competes against k opponents drawn from the current population. The scoring system awards 1 point for a win, 0 points for a loss, and 0.5 points for a draw. The cumulative score at the conclusion of the tournament serves as the chromosome's fitness value.

This approach is highly advantageous as it avoids making prior assumptions about opponent behaviour, thereby mitigating the risk of overfitting against a specific, static strategy. Because the pool of opponents dynamically evolves across generations, the evaluation metric is inherently robust. While there exists a theoretical risk of a single chromosome dominating the population and reducing genetic diversity, empirical observations during development indicated that this was not a prevalent issue.

3.3.3 Selection, Crossover, and Mutation Operators

Once fitness values are assigned, chromosomes are selected for reproduction using the Roulette Wheel Selection method [18]. The probability of a chromosome being selected is directly proportional to its fitness value divided by the aggregate fitness of the entire population. For reproduction, the arithmetic crossover method is utilized [18]. Unlike combinatorial problems (such as the Travelling Salesman Problem) where genes represent distinct categorical nodes and arithmetic operations might yield invalid states, the genes in this context represent independent numerical weights. Therefore, arithmetic crossover does not risk violating structural constraints.

Finally, to maintain genetic diversity, a mutation function is applied. At each generation, every gene within the population possesses a predefined probability of mutation. If triggered, the gene is replaced by a uniformly distributed random integer within the $[0, 50]$ interval.

3.4 Testing Approaches

3.4.1 Manual State Verification via Custom UI

To rigorously validate the underlying game engine and the Minimax decision-making logic, a diagnostic testing harness was integrated directly into the graphical user interface. By engaging a 'Free Move' developer mode, the UI transitions from a standard game client into an interactive, visual debugging environment.

A critical feature of this harness is the ability to save and load arbitrary board states. This capability facilitated deterministic scenario testing, which is essential for isolating complex edge cases in adversarial search algorithms. For instance, specific board configurations where the opponent possessed an immediate, one-turn winning threat were manually constructed and injected into the engine. By triggering the AI's search function from these controlled states, it was possible to empirically verify whether the Minimax implementation correctly identified the sole "loss evasion" moves. This visual, scenario-driven approach proved invaluable for confirming the functional correctness of the heuristic evaluation function and the alpha-beta pruning logic without relying solely on opaque console outputs.

Furthermore, a chronological state traversal mechanism navigable via UI directional controls was implemented to audit the move history of a given session. During extended adversarial testing against the AI, this feature allowed for the step-by-step playback of the executed game tree. It served as a vital validation tool to ensure that the AI strictly adhered to the mathematical invariants of the game rules, guaranteeing that no illegal state transitions were generated or executed during deep, prolonged gameplay.

3.4.2 Execution Time Profiling Methodology

To mitigate the risk of performance degradation during the optimization phase, an empirical, profiling-driven methodology was adopted. Given the hybrid C/Python architecture of the project, performance bottlenecks were inherently difficult to isolate using standard Python profilers. Consequently, `py-spy` was utilized as a sampling profiler. To ensure complete visibility into the underlying C extensions during execution, the C codebase was compiled using `gcc` with debug symbols explicitly preserved (e.g., via the `-g` flag). Post-execution, the profiling data was exported to an SVG format to generate flame graphs. This visual representation of the call stack allowed for the immediate identification of heavily localized computational bottlenecks. Code optimization was then systematically managed through the following iterative lifecycle:

1. **Bottleneck Identification:** Isolate the most computationally expensive function via flame graph analysis.
2. **Targeted Optimization:** Refactor the algorithm to reduce temporal complexity or minimize invocation frequency (e.g., via bitboard manipulation enhancements).
3. **Performance Verification:** Benchmark the targeted modification to confirm a statistically significant reduction in total execution time.
4. **Regression Testing:** Ensure the optimization maintained strict functional correctness without introducing logical regressions.

This continuous profiling loop was sustained until the Minimax algorithm achieved a stable execution time of a few seconds per move at a search depth of $d = 3$. Scaling the search space to $d = 4$ resulted in execution times exceeding 100 seconds per move. Because Minimax expands exponentially with respect to the branching factor ($O(b^d)$), further low-level optimization of the bitboard yielded diminishing returns. Therefore, a pragmatic architectural

decision was made to terminate the optimization phase. A depth of 3 provided highly satisfactory AI proficiency, representing an optimal trade-off between computational tractability and heuristic search depth within the project's time constraints.

3.4.3 Empirical Validation Of The Evolutionary Heuristic Optimization

Early considerations of the GA's tournament selection highlighted a fundamental algorithmic challenge known as the "relative fitness" problem [28]. Because a candidate chromosome's fitness is evaluated by playing matches against a limited subset of the current population, its apparent viability is entirely dependent on its immediate neighbours. For instance, a globally excellent gene tested exclusively by other highly evolved genes might yield a mediocre win-rate and be incorrectly discarded. Conversely, an average gene situated in a poorly performing sub-population might appear exceptional, an issue that can ultimately lead to a loss of evolutionary gradient or co-evolutionary disengagement [11].

To test the results of GA runs and to establish an objective performance baseline, supposedly elite chromosomes were systematically evaluated against a mostly fixed, heterogeneous benchmark pool rather than relying solely on intra-population play. This adversarial cohort was deliberately stratified into three categories:

- **Stochastic Agents:** AI utilizing uniformly randomized weight vectors to establish an absolute minimum baseline of game-playing competence.
- **Legacy Iterations:** Chromosomes extracted from intermediate generations of the GA, serving as a control group to empirically track evolutionary progress over time.
- **Domain-Expert Agents:** Heuristic profiles constructed via manual parameter tuning (hand-crafted).

By aggregating the win/loss ratios across this diverse array of opponents, the tournament yielded a robust, objective fitness metric. This comprehensive scoring mechanism established a definitive performance hierarchy, successfully isolating the globally optimal chromosome and choosing which chromosome to use in the final version of the AI.

Chapter 4

Results, Evaluation and Discussion

This chapter presents the empirical evaluation of the system across three dimensions: the throughput gain achieved by the C/Cython bitboard engine over the legacy pure-Python implementation, the effectiveness of the adversarial search optimisations described in Section 3.2.1, and the quality of the weight vector evolved by the Genetic Algorithm. Each result is interpreted against the design decisions documented in Chapters 2 and 3.

4.1 Engine Performance Evaluation

4.1.1 Legacy Python to C/Cython Migration

Table 4.1 reports the mean NPS achieved by each engine at depths 1–3 on the initial board position, with alpha-beta enabled and the transposition table disabled. This configuration isolates raw compute throughput by removing TT-initialisation variance and hit-rate variability.

Table 4.1: Mean nodes-per-second (NPS): Legacy Python vs. Cython core, alpha-beta ON, transposition table OFF, initial board position, $n = 5$. Legacy figures are estimated from evaluation-function calls; Cython figures are exact C counters.

Depth	Legacy NPS (est.)	Cython NPS (exact)	Speedup
1	24,053	1,542,371	64×
2	16,341	1,443,758	88×
3	20,660	1,495,918	72×

The Cython engine sustains 1.44–1.54 million NPS across depths 1–3, while the Legacy engine achieves 16,000–24,000 estimated NPS, placing the measured speedup in the range 64–88×. A key methodological limitation is that the Legacy figures are estimated from evaluation-function call counts rather than measured by exact hardware counters, as no instrumented node counter was present in the legacy codebase at the time of measurement. This introduces systematic uncertainty: the estimate is sensitive to alpha-beta cut behaviour and may undercount or overcount depending on how pruning distributes evaluation calls across the tree. The speedup figures should therefore be treated as order-of-magnitude characterisations rather than precise ratios. Nevertheless, a factor of 64–88× is sufficiently large that no plausible measurement error would reduce it to a negligible margin; the performance gap between the two engines is unambiguous.

The speedup is consistent with established benchmarking of Cython extensions against pure-Python equivalents in compute-bound loops [4], and with DOD studies reporting order-of-magnitude cache-efficiency gains when eliminating pointer-chasing data structures [39, 1].

4.1.2 Legacy Engine Scaling Failure at Depth 3

Table 4.2 presents the Searches per second (SPS) which is the number of full iterative-deepening sequence completed per second and mean wall-clock time per complete iterative deepening depth- d search for both engines. A critical empirical observation is that the Legacy engine failed to complete a single depth-3 search within the 100-second timeout. This is consistent with the theoretical game-tree complexity: at a branching factor of 440 (Appendix D), exhaustive enumeration to depth 3 produces approximately $440^3 \approx 85$ million leaf nodes, which at 20,000 estimated NPS would require roughly 70 minutes. Even with alpha-beta enabled, Legacy Engine real-time play at this depth is entirely infeasible.

Table 4.2: SPS and mean wall-clock time per complete iterative-deepening call: Legacy Python vs. Cython core, alpha-beta ON, transposition table OFF, initial board position, $n = 5$.

Depth	Legacy SPS	Legacy s/search	Cython SPS	Cython s/search
1	45.03	0.022	2,451	0.0004
2	1.40	0.714	57.2	0.017
3	timeout	timeout	0.08	12.4

The Cython engine, by contrast, completes a depth-1 search in under half a millisecond and a depth-3 search in less than 13 seconds without the transposition table. Section 4.2.1 demonstrates how the transposition table reduces this to under one second in steady-state play.

4.1.3 Discussion

The 64–88× NPS improvement is the product of two compounding engineering decisions rather than a single change. The primary contribution is the bitboard representation described in Section 2.1: replacing pointer-chasing Python objects with packed bit arrays processed by hardware bit-manipulation instructions yields the cache-efficiency gains characteristic of Data-Oriented Design [39, 1, 8]. The second, equally significant contribution is the migration of every performance-critical loop in the pipeline — move generation, legality checking, position evaluation, and the minimax kernel — to compiled C via the Cython bridge described in Section 2.4. This eliminates per-bytecode interpretation overhead and removes the GIL [3] as a throughput ceiling entirely. The two factors are not readily separable from the benchmark data alone, as no intermediate build retaining Python data structures but compiling only the loop bodies was produced; the measured speedup should therefore be understood as their combined effect.

4.2 Final AI Search Performance

This section evaluates the effectiveness of alpha-beta pruning and the transposition table (introduced in Chapters 1 and 3 respectively) using a saved mid-game fixture position, which presents a denser transposition profile and more representative conditions for actual gameplay than the initial position.

4.2.1 Minimax Enhancements Effectiveness

Table 4.3 records the nodes evaluated per single depth-3 search and the corresponding wall-clock time, under three configurations: no pruning, alpha-beta alone, and alpha-beta combined with the transposition table. Because unoptimised depth-3 searches take approximately 49 seconds, this benchmark ran with a 300-second safety timeout; all five samples completed without interruption. Figure 4.1 visualises the three node counts on a logarithmic scale to make the relative magnitudes immediately apparent.

AB	TT	Nodes/search	s/search	Node reduction	Time reduction
Off	Off	82,262,629	49.0	1.0×	1.0×
On	Off	21,460,422	12.8	3.8×	3.8×
On	On	≈1,147,000	≈0.73	≈72×	≈67×

Table 4.3: Nodes per single depth-3 search and mean wall-clock time per search by pruning configuration, Cython core engine, mid-game fixture position, $n = 5$. Reduction factors are relative to the no-pruning baseline. AB+TT figures represent steady-state throughput with a warm transposition table.

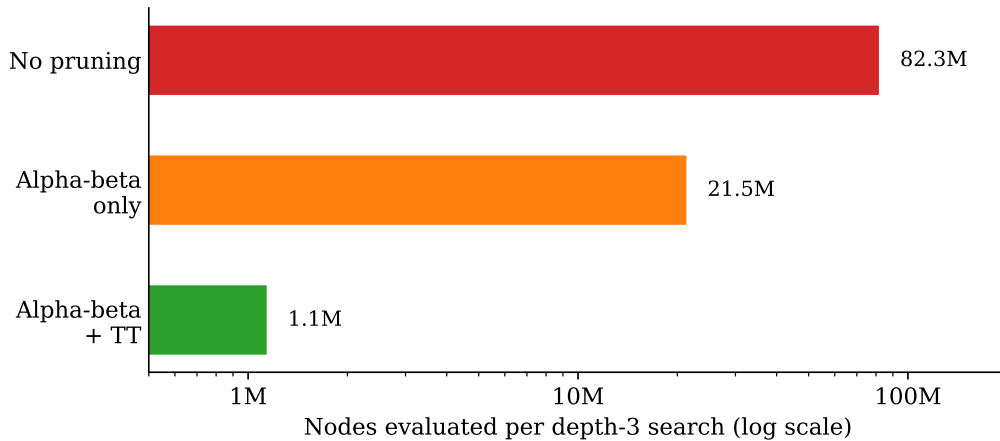


Figure 4.1: Nodes evaluated per depth-3 search under three pruning configurations. Note the logarithmic scale.

Alpha-Beta Pruning

Alpha-beta pruning alone reduces the per-search node count from 82.3 million to 21.5 million, a factor of 3.8×. As established in Chapter 1, the theoretical best-case reduction is $O(b^d) \rightarrow O(b^{d/2})$, which at $b = 440$ and $d = 3$ predicts fewer than 10,000 nodes under perfect move ordering. The empirical 3.8× reduction falls substantially short of this ideal, attributable to Nonaga’s exceptionally high branching factor producing a broad, flat position landscape at shallow depths: many moves receive similar evaluation scores, so the alpha-beta window is poorly constrained and cutoffs are less frequent than in narrower games.

Transposition Table Effectiveness

The most significant performance contribution comes from the transposition table operating in concert with iterative deepening, as described in Section 3.2.2. After the TT is populated by

the shallower passes (depths 1 and 2), the hash-move heuristic places the cached best move at the front of the child list for the depth-3 pass, substantially improving alpha-beta cutoff rates. The result is a further reduction from 21.5 million nodes (alpha-beta only) to approximately 1.15 million nodes per depth-3 search, a $19\times$ additional reduction attributable solely to the TT, and a total reduction of $\approx 72\times$ relative to brute force. The corresponding wall-clock time falls from 49 seconds to approximately 0.73 seconds, transforming depth-3 search from a computationally intractable operation into one viable for real-time interactive play.

A secondary effect is a slight decrease in raw NPS when the TT is enabled: 1.38 MNPS (alpha-beta + TT) versus 1.50 MNPS (alpha-beta only). As noted in Section 4.4.2, TT probe and write operations introduce per-node memory-access overhead. However, the reduction in total nodes visited far outweighs this per-node cost, yielding the net $67\times$ wall-clock improvement.

The TT warm-up behaviour is also evident in the Final Engine Benchmark data (Section 4.2.3): the first sample at depth 3 on the `saved_board.json` fixture evaluates 660,140 nodes and achieves 110,021 NPS with a cold TT, whereas the four subsequent samples settle into the 50,000–60,000 node range at 8,000–10,000 NPS once the table is populated.

4.2.2 Transposition Table Hit-Rate Analysis

The two previous subsections characterised pruning effectiveness at the node-count level. To complement this, a dedicated TT-metrics benchmark was executed against a later engine build (7a7b534) instrumented to expose TT statistics counters, which extends the Final Engine Benchmark harness to expose per-call TT counters: probe count, hit count broken down by bound type: EXACT, LOWER, UPPER (E/L/U) and the number of times the hash-move was both tried and provoked an immediate cutoff. Table 4.4 reports the aggregate statistics across both fixture positions at depths 1–3.

Table 4.4: Transposition table statistics per depth: mean probes, mean hits, overall hit rate (HR), bound-type breakdown, and hash-move cutoff rate (HMCR). Commit 7a7b534, both fixture positions combined, $n = 5$ samples per depth. `saved_board_backup.json` had a fully warm TT at all depths due to prior `saved_board.json` runs; cold-TT values in the final column are drawn from sample 0 of `saved_board.json` only.

D	Mean probes	Mean hits	HR	E/L/U	HMCR	Cold HR*
1	17,750	17,748	99.99%	100 / 0 / 0%	—	99.98%
2	31,629	31,176	98.57%	≈ 100 / 0 / 0%	—	91.8%
3	52,565	48,418	92.1%	97.3 / 2.4 / 0.01%	63.9%	70.4%

* Sample 0 of `saved_board.json` only (cold transposition table).

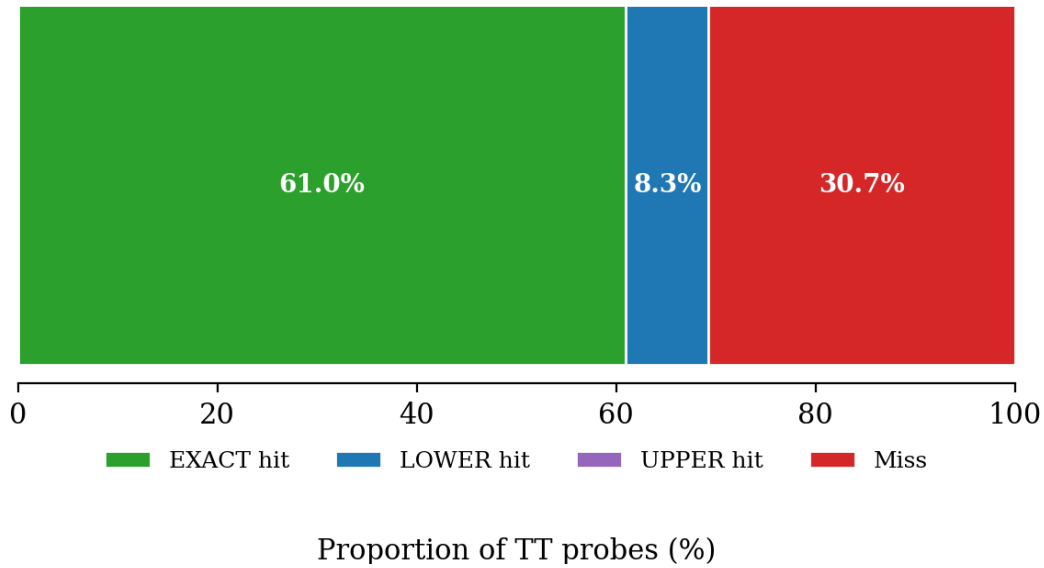


Figure 4.2: Breakdown of transposition table probe outcomes during the cold depth-3 search on `saved_board.json` (sample 0): **EXACT** hits enable immediate cutoffs, **LOWER**/**UPPER** hits tighten the α/β window, and misses proceed to full subtree expansion.

Several observations follow directly from Table 4.4 and Figure 4.2:

- First, the hit rate under warm conditions is above 90% at all depths. This means that in steady-state play, every position the engine encounters has already been evaluated and cached; the minimax kernel reduces to a sequence of hash-table lookups with no subtree expansion, which explains the very low steady-state node counts observed in Table 4.5.
- Second, the cold-start hit rate degrades sharply with depth: 99.98% at depth 1, falling to 91.8% at depth 2 and 70.4% at depth 3. This reflects the well-established iterative-deepening reuse property [36]: the depth-1 pass fills the TT with shallow evaluations; the depth-2 pass finds these entries and extends them; and by the time the depth-3 pass begins, approximately 70% of the positions it encounters are already cached. The remaining 29.6% are genuinely novel positions at the frontier of the depth-3 tree (Figure 4.2).
- Third, at depth 3 the **EXACT** bound accounts for 97.3% of all hits, with **LOWER** bounds comprising 2.4% and **UPPER** bounds less than 0.01%. The dominance of **EXACT** hits indicates that the TT mostly returns complete, definitive evaluations rather than partial window bounds, which is the strongest possible form of TT reuse as it permits an immediate return without any further search.
- Fourth, the hash-move cutoff rate (HMCR) of 63.9% at depth 3 quantifies the move-ordering contribution of the TT directly: when the hash-move is placed first in the child list, it triggers an alpha-beta cutoff in nearly two out of three cases, which is the primary mechanism by which TT and alpha-beta synergise to produce the $\approx 72\times$ total node reduction reported in Section 4.2.1.

4.2.3 Combined Optimisation: Final Engine Throughput

Table 4.5 reports the NPS and mean nodes evaluated by the final engine (8901593) with both alpha-beta and the transposition table enabled, across two saved fixture positions at depths 1–3.

Table 4.5: Final engine (Commit 8901593) throughput: NPS and mean evaluated nodes per depth, Final Engine Benchmark, $n = 5$ samples per fixture. Median NPS is reported alongside the mean at depths 2–3 due to cold-TT outlier variance in the first sample.

Fixture	Depth	Mean NPS	Median NPS	Mean nodes
saved_board.json (cold)	1	2,850	2,801	17,100
	2	9,708	5,915	58,252
	3	28,857	8,496	173,155
saved_board_backup.json (hot)	1	2,633	2,580	15,799
	2	5,660	5,557	33,962
	3	8,958	8,524	53,749

The NPS figures (2,600–28,800 depending on depth and TT warmth) in Table 4.5 are considerably lower than the figures mentioned in Section 4.2.1 (≈ 1.5 MNPS). This gap arises from two implementation differences between the two harnesses. (Discussed in more detail in Appendix F.1):

Despite the lower absolute figures relative to the Engine Improvement Benchmark harness, the NPS values confirm that the engine sustains adequate throughput under full Python orchestration, with empirical gameplay confirming depth-3 decisions consistently within 1–2 seconds.

4.2.4 Discussion

The search evaluation exposes an important architectural trade-off: raw NPS and real-game decision latency are different objectives, and the implementation correctly prioritises the latter. The combined $\approx 72\times$ node reduction falls significantly below the theoretical $b^{d/2}$ optimum, primarily because the evaluation function has limited discriminative power at shallow depths, leaving many move orderings poorly ranked. As noted in Section 3.2.2, no killer or history heuristic was implemented; the sole ordering mechanism is the TT hash-move. The literature reports that these additional heuristics routinely yield a further 30–60% node reduction [29, 5], suggesting considerable headroom for future improvement.

4.3 Genetic Algorithm Optimisation Results

This section will be completed once the current GA benchmark run concludes. The best chromosome recovered from the completed GA stages is $w = [35, 14, 40, 20, 35, 40, 39, 34]$, stored in `parameters.json`. GA metrics for all completed stages are logged in `ga_metrics.csv` and `ga_metrics_stage_1-4.csv`.

4.4 Evaluation Summary

The C/Cython bitboard engine sustains 1.44–1.54 million NPS in raw processing rate, a 64–88 $\times$ improvement over the legacy pure-Python implementation (Section 4.1.1), validating the decision to implement a custom hexagonal bitboard from first principles [8]. Because both engines execute the same algorithm at a given depth, this speedup in processing rate translates directly to an equivalent reduction in decision latency: the legacy engine required more than 100 seconds per depth-3 search; the Cython engine reduces this to approximately 12 seconds even without any caching (Section 4.1.1).

Alpha-beta pruning reduces the depth-3 node count by a factor of 3.8 $\times$ relative to brute force (Section 4.2.1). The transposition table, operating synergistically with iterative deepening, contributes a further 19 $\times$ reduction in nodes visited per search through the hash-move mechanism (Section 4.2.1), yielding a combined $\approx 72\times$ total reduction. In practice, this converts a 49-second depth-3 decision into a sub-second one under warm-cache conditions (Section 4.2.1), enabling real-time gameplay at depth 3.

Brute-force enumeration of the evaluation-function weight space is demonstrated to be computationally infeasible at any realistic scope (Appendix E), with even a severely restricted domain requiring approximately 169 days on 46-core HPC hardware. The GA results, including convergence curves and tournament validation of the evolved weight vector

$w = [35, 14, 40, 20, 35, 40, 39, 34]$ against the benchmark pool described in Section 3.4.3, will complete this chapter once the current benchmark run concludes.

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Appendix A

Self-appraisal

A.1 Critical self-evaluation

A.1.1 Project Scope and Time Management

Development proceeded more reactively than planned. Rather than defining a comprehensive feature roadmap upfront, work on each component began when it became the immediate bottleneck. This was partly a consequence of the initial Waterfall plan proving too rigid: new algorithmic questions emerged during implementation that required revisiting the background material, making a clean design-then-build separation impractical. The transition toward Agile methodologies was therefore organic, and proved more effective in this context, as short iteration cycles allowed design decisions to incorporate findings from earlier implementation phases.

The most significant scope casualty was machine learning evaluation functions of the kind employed by AlphaZero [35] and NNUE-based engines [22]. These represent the techniques that enable superhuman play and were an early ambition, but the research and experimentation they require placed them outside the feasible scope of a solo project under concurrent module workload. Deprioritising them in favour of depth in the GA and bitboard components was the correct call. Progress also stalled at points due to motivation dips, a structural vulnerability of solo projects that teams naturally absorb collectively. Overall, the final system meets all four core objectives stated in Chapter 1; the exclusion of machine learning remains the primary gap relative to the original vision.

A.2 Personal reflection and lessons learned

A.2.1 Lessons Learned in Cross-Language Integration

The decision to implement the engine in C, expose it through Cython, and orchestrate it from Python was a deliberate risk. No prior implementation existed for Nonaga’s specific topology, and the cross-language stack introduced complexity at every layer. The motivating judgement was that an ambitious but incomplete project was preferable to a safe but computationally inadequate one: the legacy Python engine could not support meaningful adversarial search, and accepting that would have constrained the entire project.

One risk that was not adequately anticipated was profiling a mixed-language codebase. Standard Python profilers cannot instrument the C kernel; C-level profilers require debug symbols and produce output difficult to correlate with Python call sites. A solution was found using `py-spy` with `gcc -g` debug symbols to generate full-stack flame graphs [12], but only after the architectural commitment had been made. Prototyping the observability toolchain before committing to the stack would have surfaced this earlier.

A second lesson concerns the Genetic Algorithm and epistasis. The working assumption was

that its effects on convergence could be managed by increasing tournament size k or generation count. This proved incorrect: the eight evaluation weights interact non-linearly, and increasing population diversity does not disentangle these dependencies. A linkage-learning approach [24, 26] would theoretically address this but was deemed out of scope, as discussed in Chapter 2. The episode illustrated the gap between understanding a concept theoretically and estimating its practical consequences correctly.

Finally, a background in chess and its engine literature shaped many implementation choices. Techniques such as iterative deepening, Zobrist hashing, and alpha-beta pruning with transposition tables were adopted partly because of their success in that domain. This provided a strong theoretical grounding but was occasionally a source of miscalibration: Nonaga’s substantially higher branching factor and shallower average game depth mean that the relative payoff of search depth versus move ordering differs considerably from chess, and this required conscious adjustment throughout.

A.3 Legal, social, ethical and professional issues

A.3.1 Legal issues

The implementation uses several open-source libraries: `pygame` (LGPL v2.1), `Cython` (Apache 2.0), `py-spy` (MIT), `setuptools` (MIT), `snakeviz` (BSD 3-Clause), and `tornado` (Apache 2.0). All licences permit non-commercial academic use without restriction. The C engine and all application logic were written from scratch.

Nonaga is a commercially licensed game [2]; its rules and branding are the intellectual property of its creator. This implementation makes no claim of originality over the rules and was developed solely for academic research. Any commercial deployment would require explicit authorisation from the rights holder. At least one other public software implementation of Nonaga exists; it was not reviewed in detail and no code or assets from it were used.

A.3.2 Social issues

Nonaga is a highly abstract two-player game with no content relating to social issues, demographic groups, or politically sensitive topics. Its social footprint is minimal. The main social consideration is accessibility: a colourblind display mode and keyboard-only navigation were not implemented, as the interface was a demonstration platform rather than a production release. These would need to be addressed before any public distribution.

A.3.3 Ethical issues

The system raises no significant ethical concerns. No user data is collected, stored, or transmitted: the application requests no username, logs no gameplay, and communicates over no network. The only persistence is a local JSON file storing the board state on explicit user request, containing no personal information.

A.3.4 Digital Game Addiction and Gambling Considerations

This concern was raised during the assessor meeting [16]. Nonaga-Online exhibits none of the design patterns associated with addictive game mechanics: no monetisation, no reward loops, no streaks, no social pressure, and no online multiplayer. Attributing addiction risk to this application would be equivalent to attributing it to a chess program; the software offers a cognitive challenge but none of the operant conditioning scaffolding characteristic of genuinely addictive products. No gambling mechanism exists, and the game’s abstract nature makes it an unlikely target for informal wagering. Digital game addiction and gambling are not applicable concerns for this project.

A.3.5 Professional issues

Version control was managed through Git with a branching workflow: each major feature was developed on a dedicated branch and merged after validation. The repository includes a README and inline comments at key algorithmic decision points. Performance validation used instrumented benchmark harnesses rather than ad hoc timing.

For a production release, several improvements would be needed: more formally specified interface contracts between the C kernel, Cython bridge, and Python layer; substantially expanded in-code documentation; and a higher-level architectural document describing inter-file dependencies. A tracked issue list would provide external contributors with clear entry points, particularly for machine learning evaluation functions and the accessibility features noted above.

Appendix B

External Material

All game logic, the C/Cython bitboard engine, the minimax AI kernel, the Genetic Algorithm framework, the benchmarking harnesses, and the PyGame graphical interface were designed and implemented from scratch as part of this project. No external code was incorporated into the system.

The following open-source libraries were used as dependencies and were not modified:

- **pygame** (v2.6.1, LGPL v2.1): graphical rendering and event handling for the user interface.
- **Cython** (v3.2.4, Apache 2.0): compilation of the `.pyx` bridge modules linking the C engine to the Python orchestration layer.
- **py-spy** (v0.4.1, MIT): sampling profiler used to generate flame graphs of the full C/Python call stack during performance analysis.
- **setuptools** (v82.0.1, MIT): build tooling for compiling the Cython extensions.
- **snakeviz** (v2.2.2, BSD 3-Clause): visualisation tool for `cProfile` output, used during early profiling stages.
- **tornado** (v6.5.4, Apache 2.0): dependency of `snakeviz`.

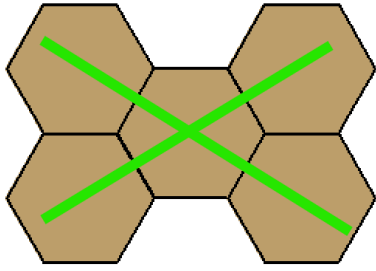
The Nonaga board game rules are the intellectual property of their creator [2] and were used solely for academic purposes. No assets, branding, or code were taken from any existing Nonaga software implementation.

Appendix C

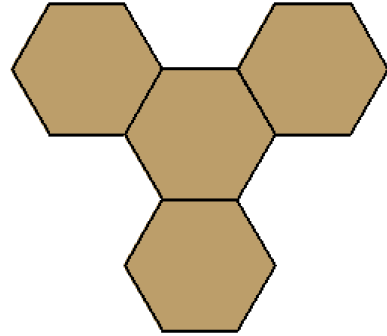
Tile Movement Legality Rule

In Nonaga, a tile may only be moved by sliding it along one of the six axial directions of the hexagonal grid, provided no neighbouring tile blocks the path. While this constraint is intuitive on a physical board, formalising it algorithmically requires a precise geometric criterion.

A tile is unmovable if and only if its immediate neighbours form one of two blocking configurations: an X-shaped arrangement (Figure C.1a) or a triangular arrangement (Figure C.1b). Any rotation of either pattern preserves the blocking property. If no subset of a tile's immediate neighbours matches either configuration, the tile is freely slidable. It follows that a tile with fewer than three neighbours is always movable, while a tile with five or six neighbours is always unmovable.



(a) X-shaped blocking configuration.



(b) Triangular blocking configuration.

Figure C.1: The two neighbour configurations that render the central tile unmovable. Any rotation of either pattern is equally blocking.

Appendix D

Nonaga's Branching Factor Calculation

The number of possible moves in any Nonaga board state is given by the following equation:

$$B = \sum_{i=1}^3 \sum_{j=1}^{P_i} \sum_{k=1}^{T_{i,j}} D_{i,j,k} \quad (\text{D.1})$$

Where:

- B : The total branching factor (number of possible complete turns from the current state).
- i : The index of the current player's piece ($i \in 1, 2, 3$).
- P_i : The total number of valid destinations for piece i .
- j : The index of a specific destination for piece i ($j = 1, 2, \dots, P_i$).
- $T_{i,j}$: The number of legally movable tiles, dynamically evaluated *after* piece i has been moved to destination j .
- k : The index of a specific movable tile ($k = 1, 2, \dots, T_{i,j}$).
- $D_{i,j,k}$: The number of valid placement destinations for tile k , given the board configuration resulting from move j .

To determine the initial branching factor, the calculation can be simplified by exploiting the symmetrical nature of the board on all 3 axis of the central hexagon. First, $D_{i,j,k}$ is the same for all i, j, k . Second, calculating the branching factor for only one piece and then multiplying the result by three results in the branching factor of the position. Third, for each piece two moves yield the same number of movable tiles.

Calculating the number of tile moves for 2 piece moves, multiplying one of them by 2 and adding it to the other results in all possible piece and tile moves for one piece and multiplying that number by three results in the branching factor of the position.

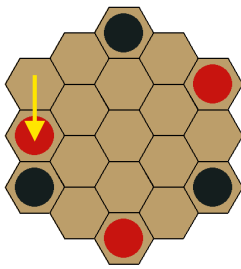


Figure D.1: Piece move selected.

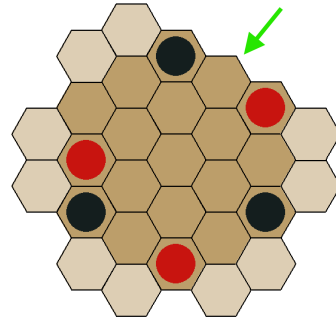


Figure D.2: Possible tile moves after the piece move in Figure D.1 of the tile pointed to

After the piece move in Figure D.1, the total number of possible tile moves, as seen in Figure D.2, is $NumOfTiles * D = 6 * 10 = 60$. It is identical to the total number of possible tile moves after the piece move in Figure D.3. It is doubled to get the number of possible tile moves for both moves.

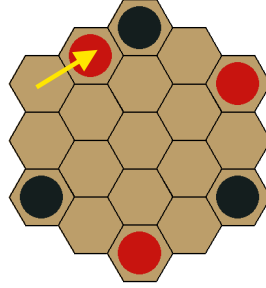


Figure D.3: The piece move with the same number of tile moves as the one in Figure D.1

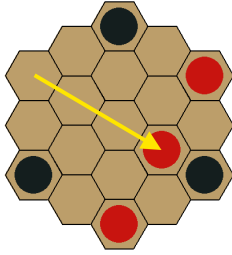


Figure D.4: The last piece move to consider

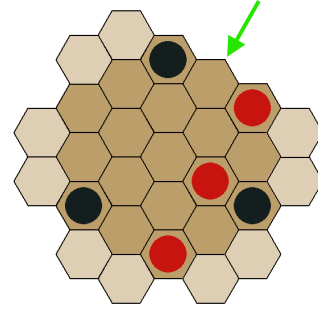


Figure D.5: Possible tile moves after the piece move in Figure D.4 of the tile pointed to

After the piece move shown in Figure D.4, the total number of possible tile moves (Figure D.5) is calculated as $N_{\text{tiles}} \times D = 7 \times 10 = 70$. Consequently, the total number of valid moves for the starting position is $3 \times (60 \times 2 + 70) = 570$. While the starting position provides a theoretical maximum, empirical analysis of middle-game states indicates an effective branching factor (b) of approximately 440. This reduction occurs because pieces become tightly packed as the game progresses, physically restricting their valid movement vectors. A branching factor of 440 is exceptionally high. For context, Chess has an average branching factor of approximately 35 [29], and Go, a game long considered a grand challenge for AI until the advent of AlphaGo in 2016 [21], averages around 250. However, evaluating a game's computational difficulty requires analysing the overall game-tree complexity, approximated as $O(b^d)$, where d is the average game depth. While Nonaga's branching factor is roughly 12.5 times larger than that of Chess, a typical game of Nonaga lasts for only about 15 moves. In contrast, Chess averages around 40 full moves (80 plies) and Go extends to roughly 150 moves. Therefore, despite the dense move generation at any individual node, the shallow depth of Nonaga significantly truncates the overall search space. This specific structural characteristic makes the game highly suitable for the Iterative Deepening Minimax architecture with Alpha-Beta pruning implemented in this project, as terminal or highly-evaluable states can be reached within a computationally feasible horizon.

Appendix E

Brute-Force Feasibility Analysis

A suggestion raised development [16] was to evaluate whether exhaustive enumeration of the evaluation-function weight space, potentially parallelised on the University of Leeds AIRE HPC cluster, could have served as a viable alternative to the GA. This section discusses the computational and theoretical infeasibility of that approach.

The evaluation function is parameterised by eight integer weights $w_i \in [0, 50]$, yielding a search space of $51^8 \approx 1.885 \times 10^{13}$ distinct vectors. Even under a severely restricted apex-gene formulation capping each weight at $[1, 6]$, the candidate space is $6^8 = 1,679,616$. Evaluating each candidate requires playing it against at least one opponent. Using the 20-game tournament protocol employed by the GA and a representative game duration of 20 seconds, the total sequential compute time is:

$$1,679,616 \times 20 \text{ games} \times 20 \text{ s/game} \approx 672 \times 10^6 \text{ s} \approx 21 \text{ years.}$$

Parallelising across the 46 available AIRE cores reduces this to approximately 169 days, still far exceeding both the project timeline and the available HPC allocation. For the full $[0, 50]^8$ domain the figure is larger by a factor of $(51/6)^8 \approx 11,200$, rendering the problem cosmologically infeasible in any unparallelled regime.

Beyond raw compute time, there is a deeper methodological objection: as discussed in Section 3.4.3, a chromosome’s fitness is inherently relative to the quality of its opponents [28]. Evaluating each candidate against a fixed reference agent would require selecting that reference agent non-trivially, and a poorly chosen reference would inflate or deflate the fitness of entire candidate subspaces. The GA sidesteps this by co-evolving a diverse population, which constitutes the principled and well-established solution to the relative-fitness challenge in adversarial game playing [11].

The brute-force analysis therefore confirms that the GA is not merely a pragmatic shortcut but the methodologically appropriate approach to this problem class, delivering a practical convergence in hours rather than centuries while managing the relative-fitness challenge inherent to game-theoretic parameter optimisation.

Appendix F

Benchmark Implementation

F.1 Benchmarking Harnesses Variations

First, the Final Engine Benchmark wraps each search call in Python orchestration code (game-state management, move selection, and logging) so the measured time includes Python overhead that the Engine Improvement Benchmark excludes by timing only the inner C minimax kernel.

Second, the Final Engine Benchmark accumulates nodes across all iterative-deepening passes (depths 1 through d); the shallow passes contribute many rapid, low-node calls to the elapsed-time denominator while contributing relatively few nodes to the numerator, which suppresses the reported NPS.